

AJ (Aryamaan) Jena

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Education

UCLA Anderson School of Management

PhD in Finance; Fink Center Award, Anderson Departmental Fellowship

• Coursework: Asset Pricing Theory, Empirical Asset Pricing, Econometrics, Regression Analysis, Machine Learning Methods

University of Chicago

MS in Financial Mathematics (Financial Data Science); ASEAN Merit Scholarship

SP Jain School of Global Management

BS in Data Science; Merit Scholarship; Dean's List (4 times)

Los Angeles, CA

2025–2030 (expected)

Chicago, IL

2021–2022

Mumbai and Sydney

2018–2021

Research

Conditioning or Orientation? Shrinkage Targets for Characteristic-Managed Portfolios

Working paper, 2026

- Showed with an eigenvector rotation test and permutation null that the out-of-sample gain from a characteristic-based shrinkage target comes mostly from conditioning, not characteristic information (132 portfolios, 1963–2025).
- Benchmarked 23 estimators over 575 out-of-sample months: characteristic and scaled-identity targets tie for first by Sharpe ratio, significantly ahead of RIE, POET, RP-PCA and nonlinear shrinkage after Bonferroni correction.

Returns on Presale Housing Markets in India

Working paper, 2025

- Found nominal presale returns of 2–5% a year, earned mostly at launch, with no premium for holding to completion.

Research Experience

Indian Institute of Management Bangalore

Research Associate, Prof. Venkatesh Panchapagesan

- Built presale return indices by construction stage, builder reputation and housing segment across four cities.

Harvard Business School

Research Associate, Profs. Victoria Ivashina and Josh Lerner

- Simulated mean-variance and Black-Litterman portfolios to estimate the cost of illiquidity in private equity and real estate.
- Built a panel of US financial firms; estimated the effect of acquisitions on FinTech innovation with synthetic control.

University of Chicago Booth School of Business

Research Assistant, Prof. Dacheng Xiu

- Predicted errors in Zillow's home price estimates with linear, regularized and nonparametric ML regressions.
- Built LendingClub default classifiers that became assignments and lecture material for Booth MBA students.

Industry Experience

Cboe Global Markets, Cboe Labs

Derivatives Strategy Analyst Intern; received a full-time return offer

- Developed a quote filter for the VIX calculation (recomputed every 15 seconds) that uses put-call parity bounds to flag anomalous option quotes behind index spikes; shipped as an object-oriented Python module in the firm-wide quant library.
- Built a Python SEC EDGAR library that maps tickers, CUSIPs and SIC codes to one firm ID and parses 13F holdings; turned SEC/CFTC index eligibility rules into automated tests for iBoxx indices.
- Contributed unit tests and code reviews to the shared library (Jenkins, Kubernetes); migrated team workflows to AWS.

Pareto Frontier Capital Holdings

Quantitative Research Intern; feature engineering and PCA for crypto trading signals

Chicago, IL

Oct 2021 – Jan 2022

Teaching

UCLA Anderson, TA: Credit Markets (Prof. Pénasse); Venture Capital and Private Equity (Prof. Garmaise)

Chicago Booth, TA: Financial Econometrics (Prof. Russell); Decoding FinTech (Prof. Xiu)

Fall 2026

2022

Projects

Prediction Market Price Discovery

2026

- Collected 640,000+ Polymarket and Kalshi markets; matched 759 identical contracts with TF-IDF text similarity.
- Price gaps close with a half-life of about 2 hours; Polymarket leads price discovery (Gonzalo-Granger share about 0.7).

Skills

Statistics: Shrinkage estimation, factor models, PCA, regularization, gradient boosting, Bayesian inference, synthetic control

Finance: Empirical asset pricing, portfolio construction (mean-variance, Black-Litterman), credit risk, prediction markets

Tools: Python (pandas, scikit-learn, statsmodels, LightGBM), R, SQL, Git, AWS, SGE cluster, WRDS, EDGAR